PROBABILITY AND STATISTICS

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Continuous Random Variables

- **Continuous Sample Space**
- A continuous set of numbers, sometimes referred to as an interval, contains all of the real numbers between two limits.

1. (x1, x2) is the open interval defined as all numbers between x 1 and x2 but not including either x1 or x2. Formally, $(x1, x2) = \{x|x1 < x < x2\}$.
2. [x1, x2] is the closed interval defined as all numbers between x 1 and x2 including both x1 and x2. Formally $[x1, x2] = \{x|x1 \leq x \leq x2\}$.
3. [x1, x2) is the interval defined as all numbers between x 1 and x2 including x1 but not including x2. Formally, $[x1, x2) = \{x|x1 \leq x < x2\}$.
4. (x1, x2] is the interval defined as all numbers between x 1 and x2 including x2 but not including x1. Formally, $(x1, x2] = \{x|x1 < x \leq x2\}$.

Continuous Random Variables

- Many experiments lead to random variables with a range that is a continuous interval.
- Examples:
- Measuring T , the arrival time of a particle ($S_T = \{t|0 \leq t < \infty\}$)
- Measuring V, the voltage across a resistor ($S_V = \{v| -\infty < v < \infty\}$)
- Measuring the phase angle A of a sinusoidal radio wave ($S_A = \{a|0 \leq a < 2\pi\}$).

Continuous Random Variables

- **Definition:** Cumulative Distribution Function (CDF)
- The cumulative distribution function (CDF) of random variable X is

$$F_X\,(x) \;=\; P\,[X \;\leq\; x]\,.$$

- Graphs of all cumulative distribution functions start at zero on the left and end at one on the right. All are nondecreasing, and, most importantly, the probability that the random variable is in an interval is the difference in the CDF evaluated at the ends of the interval.
- **Theorem:** For any random variable X,
- $(a)\; F_X\,(-\infty) \;=\; 0$
- $(b)\; F_X(\infty) \;=\; 1$
- $(c)\; P[x_1 < \; X \;\leq x_2] \;= F_X(x_2) \;- F_X(x_1)$

Probability Density Function

- **Definition:** Continuous Random Variable
- X is a continuous random variable if the CDF $F_X(x)$ is a continuous function.

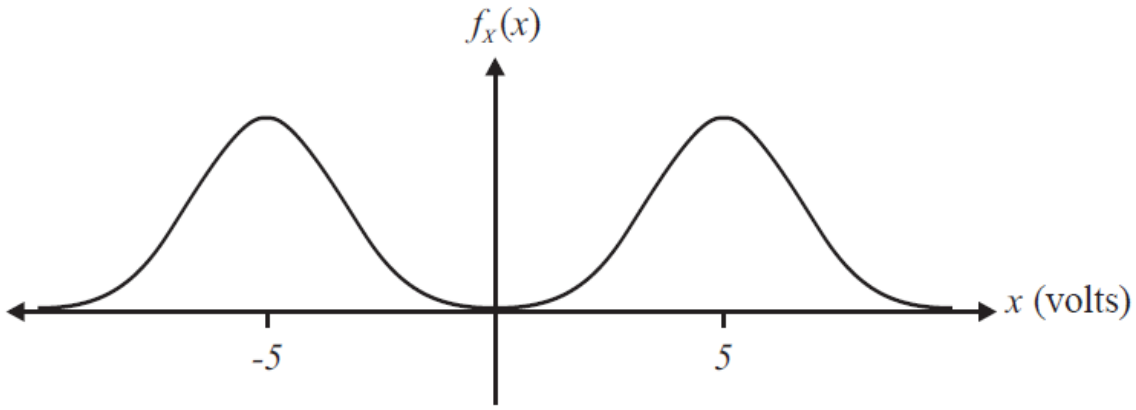
- **Probability Density Function (PDF)**
- The slope of the CDF contains the most interesting information about a continuous random variable. The slope at any point x indicates the probability that X is *near* x.
- Definition: Probability Density Function (PDF)
- The probability density function (PDF) of a continuous random variable X is:

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Probability Density Function

- The PDF is a complete probability model of a continuous random variable. While there are other functions that also provide complete models the **PDF is the most useful**. One reason for this is that the graph of the PDF provides a good indication of the likely values of observations.

- Example: Figure below depicts the PDF of a random variable X that describes the voltage at the receiver in a modem. What are probable values of X?



Probability Density Function

- Note that there are two places where the PDF has high values and that it is low elsewhere. The PDF indicates that the random variable is likely to be near −5 V (corresponding to the symbol 0 transmitted) and near +5 V (corresponding to a 1 transmitted). Values far from ±5 V (due to strong distortion) are possible but much less likely.
- **Theorem:** For a continuous random variable X with PDF $f_X\ (x)$,

1. $f_X(x) \geq 0 \ \ for\ all\ x$
2. $F_X(x) = \int_{-\infty}^{x} f_X(u)du$
3. $\int_{-\infty}^{+\infty} f_X(x)dx = 1$

Probability Density Function

- **Example 1:**

The cumulative distribution function of the random variable Y is

$$F_Y(y) = \begin{cases} 0 & y < 0, \\ y/4 & 0 \le y \le 4, \\ 1 & y > 4. \end{cases}$$

Sketch the CDF of Y and calculate the following probabilities:

(1) $P[Y \le -1]$

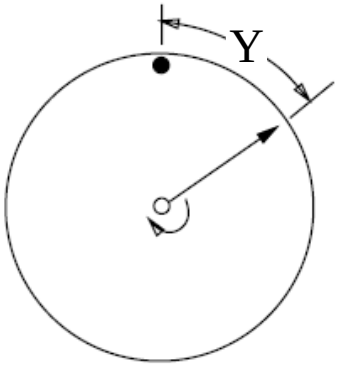
(2) $P[Y \le 1]$

(3) $P[2 < Y \le 3]$

(4) $P[Y > 1.5]$

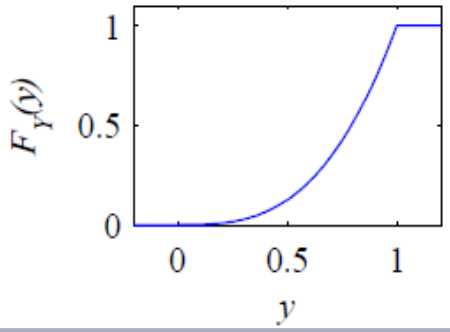
Probability Density Function

- **Example 2:** Consider an experiment and spin the pointer three times and observing Y meters, the maximum value of Y in the three spins.



- $F_Y(y) = \begin{cases} 0 & y < 0 \\ y^3 & 0 \le y < 1 \\ 1 & y \ge 1 \end{cases}$

- Find the PDF of Y and the probability that Y is between 1/4 and 3/4.



Probability Density Function

- **Example 3:**

Random variable X has probability density function

$$f_X(x) = \begin{cases} cxe^{-x/2} & x \geq 0, \\ 0 & \textit{otherwise}. \end{cases}$$

Sketch the PDF and find the following:

(1) the constant c

(2) the CDF $F_X(x)$

(3) $P[0 \leq X \leq 4]$

(4) $P[-2 \leq X \leq 2]$

Expected value and Variance

- Definition: Expected Value
- The expected value of a continuous random variable X is:

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx$$

- Theorem: For any random variable X,
- $(a)\ E[X - \mu_X] = 0$
- $(b)\ E[aX + b] = aE[X] + b$
- $(c)\ Var[X] = E[X^2] - \mu_X^2$
- $(d)\ Var[aX + b] = a^2\, Var[X]$.

Expected value and Variance

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x)dx$$

$$Var[X] = \int_{-\infty}^{+\infty} (x - \mu_X)^2 f_X(x)dx$$

- **Example:** Find the variance of the pervious examples?